



Imbalancing Techniques



Data
Science

What is Data Imbalancing?

- Imbalanced data refers to those type of datasets where the target class has a unequal distribution of observation.
- i.e. one class label has a very high number of observations and the other has a very low number of observations.



Imbalancing Techniques

1. Oversampling –

```
from imblearn.over_sampling import RandomOverSampler  
ro=RandomOverSampler()  
x_data,y_data=ro.fit_resample(x,y)
```

2. Undersampling

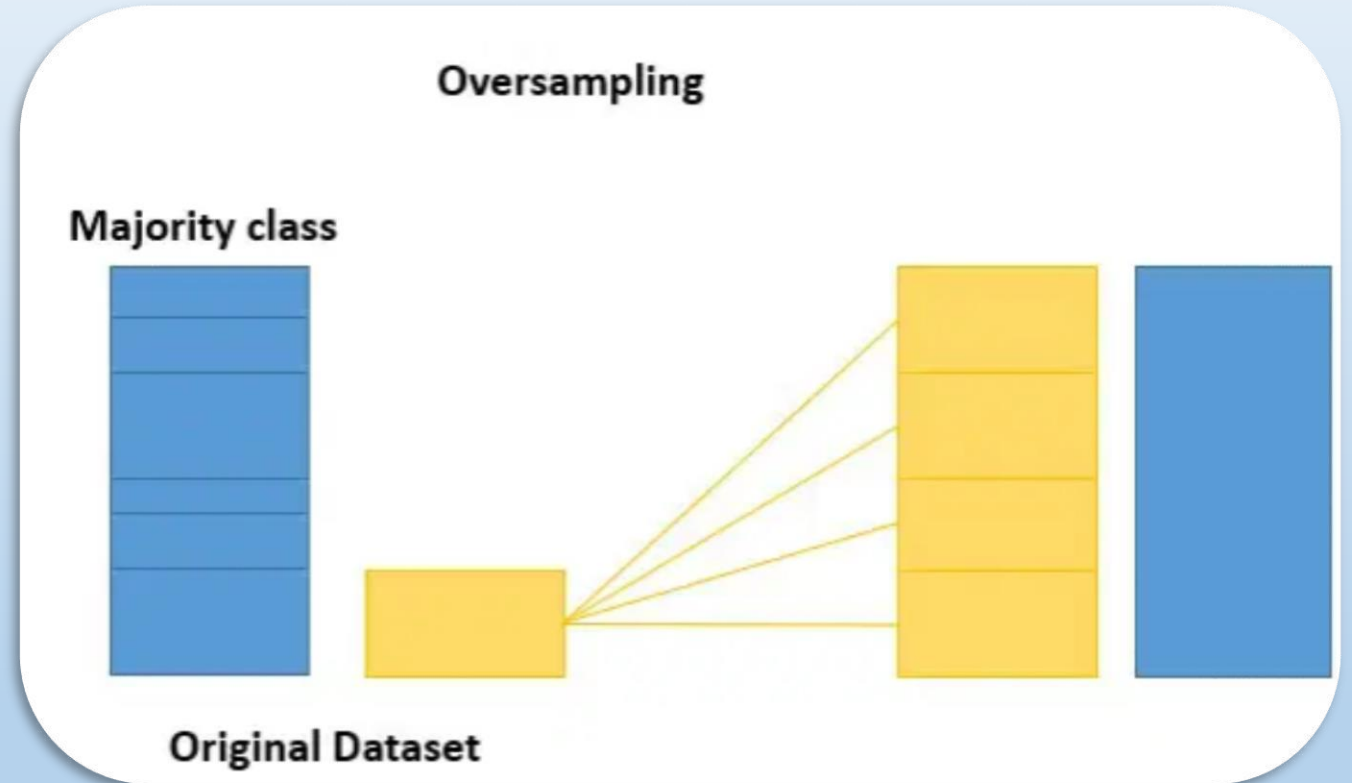
```
from imblearn.under_sampling import NearMiss  
nm=NearMiss()  
x1_data,y1_data=nm.fit_resample(x,y)
```

3. SMOTE –

```
from imblearn.over_sampling import SMOTE  
s=SMOTE()  
x2_data,y2_data=s.fit_resample(x,y)
```

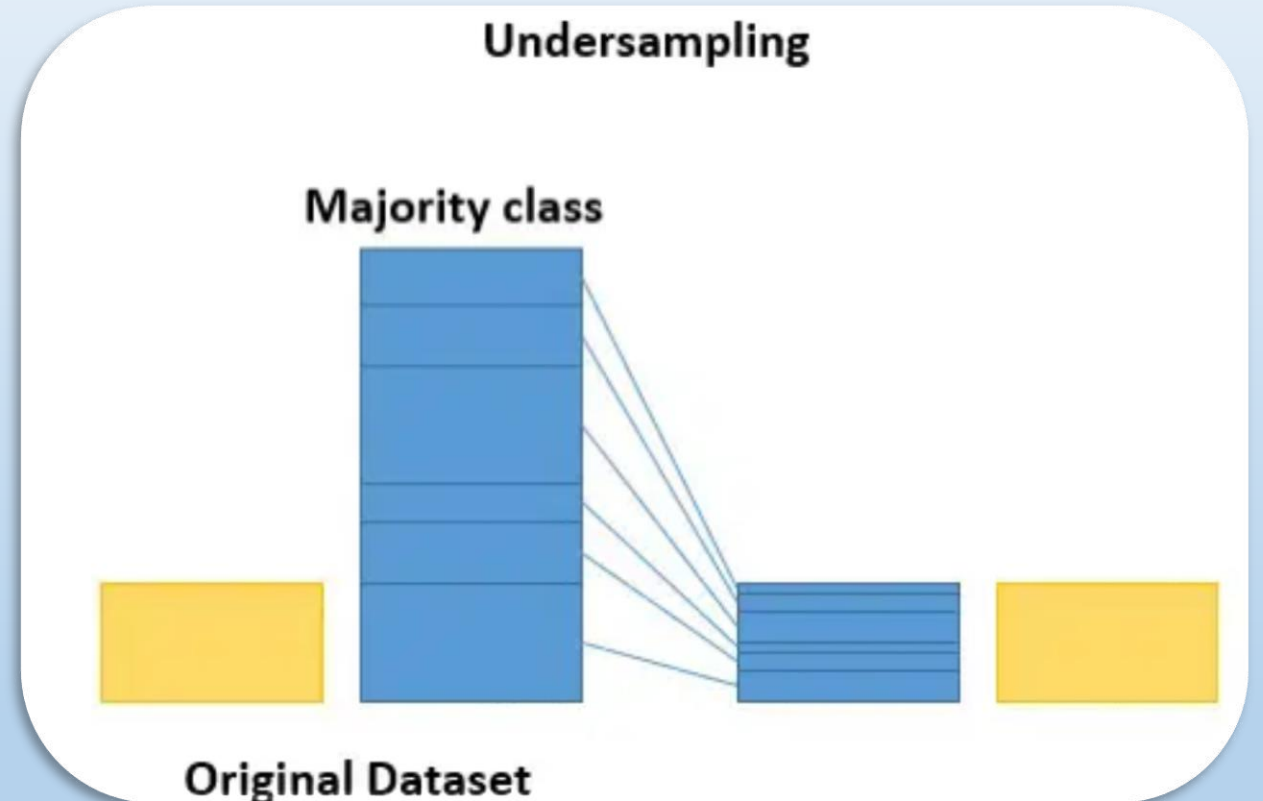
Oversampling

1. Oversampling is a imbalancing technique.
2. It is used for binary as well as multi-classification categorical data.
3. This technique is used to modify the unequal data classes to create balanced data.
4. It increases the value minority to majority in the dependent variable.
5. It takes input values randomly in the given dataset.



Undersampling

1. Random Under sampling is a imbalancing technique.
2. It is used for binary as well as multi-classification categorical data.
3. This technique is used to modify the unequal data classes to create balanced data.
4. It decreases the value majority to minority in the dependent variable.
5. It takes input features values randomly in the given dataset.
6. This is done until majority and minority class instances are balanced out.



Synthetic Minority Oversampling Technique (SMOTE)

1. SMOTE is a imbalancing technique.
2. It is used for binary as well as multi-classification categorical data.
3. This technique is used to modify the unequal data classes to create balanced data.
4. It increases the value minority to majority in the dependent variable.
5. In SMOTE the synthetic samples are generated in the minority class.



Thank You...

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